



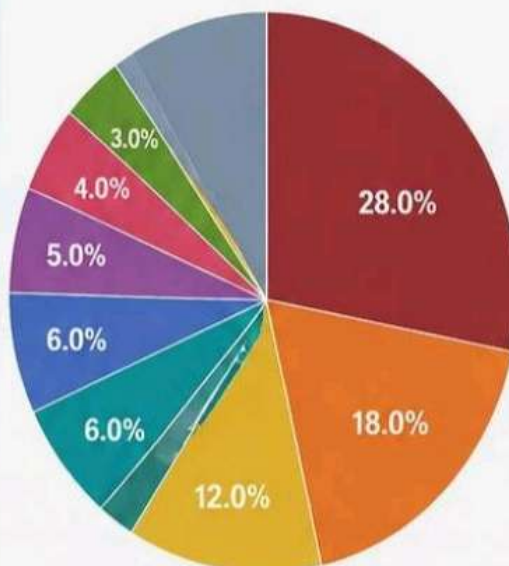
METAL CONTAMINANTS IN DRINKING WATER

MINING-AFFECTED REGIONS OF JHARKHAND, INDIA

Based on reported studies from Jharkhand mining regions (Groundwater & Surface Water)



Relative Composition of Metals in Mining-Affected Drinking Water



Iron (Fe) - 28.0%
Manganese (Mn) - 18.0%
Zinc (Zn) - 12.0%
Lead (Pb) - 8.0%
Copper (Cu) - 6.0%
Nickel (Ni) - 6.0%
Chromium [Cr(VI)] - 5.0%
Arsenic (As) - 4.0%
Cadmium (Cd) - 3.0%
Aluminium (Al) - 2.0%
Others (Co, Se, Ba, V etc.) - 8.0%

Note: Percentages represent relative contribution based on reported concentrations in various studies from Jharkhand mining regions. Actual values vary by location and season.



IRON (Fe) 28.0%

Very common in mining-affected water. High levels cause color, taste, and staining.

Typical Sources: Iron ore mining, acid mine drainage



MANGANESE (Mn) 18.0%

Common in groundwater. Affects taste and can cause staining.

Typical Sources: Mining runoff, weathering of rocks



ZINC (Zn) 12.0%

Frequently detected in mining and industrial areas.

Typical Sources: Mine drainage, ore processing



LEAD (Pb) 8.0%

Toxic metal even at low concentrations. Affects nervous system and brain.

Typical Sources: Mining activities, industrial waste



COPPER (Cu) 6.0%

From copper mining and associated industries.

Typical Sources: Mining, smelting, industrial discharge



NICKEL (Ni) 6.0%

Associated with Cr and Fe minerals.

Typical Sources: Mining, serpentinite rocks, industrial discharge



CHROMIUM [Cr(VI)] 5.0%

Hexavalent chromium is highly toxic and carcinogenic.

Typical Sources: Chromite mining, industrial effluents



ARSENIC (As) 4.0%

Naturally occurring and mining-induced. A major health risk.

Typical Sources: Geogenic sources, mining activities



CADMIUM (Cd) 3.0%

Highly toxic even in small amounts.

Typical Sources: Mining waste, battery industries



ALUMINIUM (Al) 2.0%

Common in coal mining areas.

Typical Sources: Coal mining, industrial discharge

OTHER TRACE METALS (Combined - 8.0%)

COBALT (Co)



SELENIUM (Se)



BARIUM (Ba)



VANADIUM (V)



OTHERS



Sources: Specific mining activities, geological formations, industrial processes.

HEALTH IMPACTS OF LONG-TERM EXPOSURE



Arsenic (As): Skin lesions, cancer (skin, bladder, lung), cardiovascular disease, diabetes, neurological effects.



Chromium [Cr(VI)]: Carcinogenic, causes gastrointestinal and systemic effects.



Lead (Pb): Affects brain, kidneys, and nervous system. Especially harmful to children.



Cadmium (Cd): Kidney damage, bone demineralization, and other chronic effects.



WHO GUIDELINES (Drinking Water)

- Arsenic (As): 10 µg/L (0.01 mg/L)
- Lead (Pb): 10 µg/L (0.01 mg/L)
- Chromium [Cr(VI)]: 50 µg/L (0.05 mg/L)
- Cadmium (Cd): 3 µg/L (0.003 mg/L)



IMPORTANT NOTE

These contaminants are documented in groundwater, surface water, or mine water in specific mining-affected regions of Jharkhand. Concentrations vary by geology, mining activity, and location.



SOURCE OF DATA

Compiled from various research studies conducted in Jharkhand mining regions (East Singhbhum, West Singhbhum, Dhanbad, Bokaro, Ramgarh, etc.) (2015 - 2026)

JHARKHAND MINING-REGION DRINKING-WATER FILTRATION SYSTEM

For Individual Houses

Three-stage heavy-metal treatment system designed for groundwater in Jharkhand mining regions to provide safe, clean & healthy drinking water.



3-STAGE HEAVY-METAL TREATMENT FLOW



Note: Chromium (Cr VI) occurs as an anionic species (chromate/dichromate) and is not effectively removed by the chelating resin. If Cr(VI) is detected in high levels, a separate Cr(VI)-specific treatment stage (anion exchange / reduction + filtration) should be added after testing.

COMMON CONTAMINANTS IN JHARKHAND MINING-REGION WATER



These contaminants may occur due to mining activity, weathering of minerals, and groundwater interaction.

HOW THE SYSTEM WORKS

- 1 Water passes through DMI-65® media → removes iron & manganese.
 - 2 Water flows through Purolite™ S930EPlus resin → binds and removes dissolved heavy-metal cations.
 - 3 Water passes through BAYOXIDE® E 33 HC media → adsorbs arsenic (As III & As V).
 - 4 Treated water is monitored by the outlet metal sensor for real-time safety assurance.
- Provides safe, clean & healthy drinking water for your family.

KEY FEATURES

- Compact & easy to install for individual houses
- Targets major heavy metals & arsenic in mining-affected groundwater
- Real-time monitoring with outlet sensor
- Low maintenance & long cartridge life
- Improves health & prevents long-term diseases caused by heavy-metal exposure

HEALTH RISKS OF CONTAMINATED WATER

- Arsenic (As): Skin lesions, cancer, diabetes, nerve & heart problems
- Lead (Pb): Brain & kidney damage, affects children growth
- Cadmium (Cd): Kidney damage, bone weakness
- Chromium (Cr VI): Cancer risk, lung & stomach problems
- Iron & Manganese: Staining, bad taste, affects plumbing & appliances



RECOMMENDED MAINTENANCE

- Replace cartridges as per water quality & usage.
- Regularly check outlet sensor readings.
- Get water tested periodically for best performance.



SUITABLE FOR

Individual Houses
Borewell / Groundwater
in Mining-Affected Areas
of Jharkhand



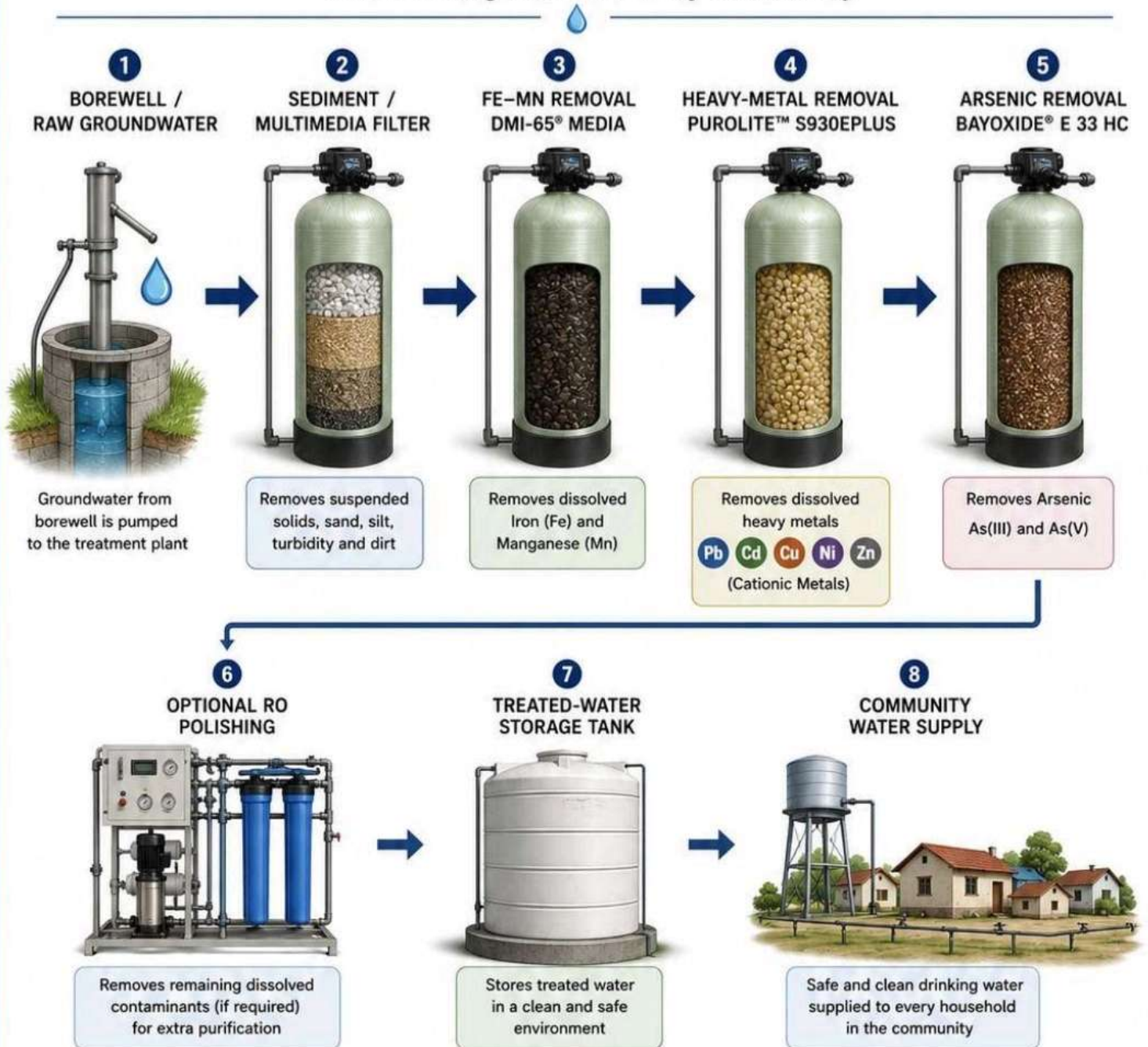
OUR GOAL

Safe Water. Healthy Life.
Stronger Families.
Stronger Jharkhand.



JHARKHAND COMMUNITY GROUNDWATER FILTRATION SYSTEM

Safe Drinking Water for Every Community



TARGET CONTAMINANTS

Fe Iron	Cu Copper	Cr(VI) Cr(VI)
Mn Manganese	Ni Nickel	Chromium (VI) requires separate treatment if detected
Pb Lead	Zn Zinc	
Cd Cadmium	As Arsenic	

KEY FEATURES

- Designed for mining-affected groundwater in Jharkhand
- Multi-stage treatment for effective removal of metals and arsenic
- Backwashable / Regenerable media vessels
- Continuous monitoring ensures safe drinking water
- Centralized system for community benefit and long-term sustainability



CLEAN WATER
HEALTHY COMMUNITY
STRONGER JHARKHAND



DESIGN BASED ON LOCAL BOREWELL TESTING

Flow rate • Daily demand • Contaminant concentration • Media capacity • Contact time

Test → Treat → Monitor → Supply



JHARKHAND MINING-REGION GROUNDWATER FILTRATION SYSTEM

SAFE DRINKING WATER FOR EVERY COMMUNITY

Three-Stage Selective Metal Treatment • Monitor • Alert • Supply

OUR GOAL



From Mining-Affected
Groundwater to
Safe, Healthy &
Sustainable
Community Water



A. HOUSEHOLD FILTRATION UNIT (For Individual House)

S.No.	Component	Specification / Purpose	Prototype Cost (₹)
1	Raw-water tank	100–200 L	700
2	Sediment pre-filter	5 µm PP cartridge + housing	300
3	Fe/Mn cartridge	DMI-65® media	800
4	Heavy-metal cartridge	Purolite™ S930EPlus resin	1,200
5	Arsenic cartridge	BAYOXIDE® E 33 HC media	1,200
6	RO stage	75–100 GPD RO membrane	900
7	pH sensor	Online pH sensor	600
8	Turbidity sensor	Online turbidity sensor	400
9	EC/TDS sensor	EC/TDS sensor module	600
10	Flow sensor	Water-flow sensor	250
11	Controller	ESP32 microcontroller	350
12	Relay + buzzer + LEDs	Control & alarm system	250
13	Pump	Small DC water pump	700
14	Solar system	Small solar panel + controller	1,200
15	Battery	12 V battery	800
16	Pipes, valves, enclosure	Plumbing, fittings, housing	700

TOTAL PROTOTYPE COST (HOUSEHOLD UNIT)

₹15,000

* Costs are prototype-scale estimates for budgeting purpose only.
Actual prices may vary by supplier, quantity and region.

TARGET CONTAMINANTS IN JHARKHAND MINING-REGION WATER



Concentrations vary by location, geology, mining activity & season.
Treatment is based on actual water test results.

IMPORTANT NOTE

- Media selection and quantity must be based on raw-water test results.
- Different villages may require different treatment capacities and media loading.
- Cr(VI) occurs as an anionic species (chromate/dichromate) and requires a separate anion-specific treatment stage if present in high levels.
- Regular monitoring and maintenance are essential for safe water.

TEST → TREAT → MONITOR → SUPPLY



These are preliminary budgetary estimates for prototype systems based on current market information (May 2025).
Actual costs will vary depending on exact capacity, raw-water quality, plant configuration, vendor quotation and site conditions.

B. COMMUNITY FILTRATION SYSTEM (1,000 LPH) (Village / Community Level)

S.No.	Component	Specification / Purpose	Prototype Cost (₹)
1	Raw-water storage tank	2,000–5,000 L	20,000
2	Sediment / multimedia filter	FRP vessel + sand/multimedia	20,000
3	Fe/Mn treatment vessel	FRP vessel + DMI-65® media	30,000
4	Heavy-metal vessel	FRP vessel + Purolite™ S930EPlus	50,000
5	Arsenic vessel	FRP vessel + BAYOXIDE® E 33 HC	40,000
6	Optional RO plant	1,000 LPH commercial RO	1,80,000
7	UV / final filtration	UV disinfection + final filter	15,000
8	Treated-water tank	5,000–10,000 L	30,000
9	Raw-water pump	Groundwater extraction pump	15,000
10	High-pressure pump	RO high-pressure pump	25,000
11	Transfer pump	Treated-water transfer pump	10,000
12	Pipes & valves	UPVC/CPVC + valves + gauges	40,000
13	Electrical / control panel	MCB, relays, contactors, etc.	30,000
14	pH sensor	Industrial online pH	5,000
15	EC/TDS sensor	Industrial EC/TDS	5,000
16	Turbidity sensor	Industrial online turbidity	5,000
17	Flow meter	Industrial flow meter	5,000
18	Pressure sensor / switch	Filter / RO pressure monitoring	5,000
19	Tank level sensor	Raw & treated water level	3,000
20	Controller	Industrial PLC / ESP32 gateway	15,000
21	GSM / 4G module	Remote monitoring & alerts	5,000
22	DTMF decoder	MT8870 DTMF decoder	200
23	VODASPEC analyzer	Online multi-element metal analyzer	TBD (OEM Quotation)
24	Installation & plumbing	Civil, electrical, plumbing works	60,000
25	Commissioning & training	Testing, startup & training	15,000

TOTAL PROTOTYPE COST (COMMUNITY SYSTEM)

₹6,70,000 +
VODASPEC (OEM Quotation)

WHY THIS SYSTEM?

- ✓ Three selective metal-treatment stages (not one-size-fits-all cartridges)
- ✓ Designed for mining-affected groundwater in Jharkhand
- ✓ Real-time monitoring with online sensors & VODASPEC analyzer
- ✓ Remote alert & control using GSM + DTMF
- ✓ Scalable, maintainable & suitable for community use

REMOTE CONTROL (DTMF)



- 1 – System status
- 2 – Start purification
- 3 – Stop purification
- 4 – Start backwash
- 5 – Check tank level
- 6 – Check filter status
- 9 – Emergency shutdown

REFERENCE (Indicative)

- DMI-65® media available in India ~ ₹350/kg (supplier listing)
- Purolite™ S930EPlus – potable water heavy-metal resin (price on request)
- BAYOXIDE® E 33 HC – arsenic removal media, NSF 61 approved (price on request)
- 1,000 LPH RO plant ~ ₹1.8 to ₹3.0 lakh (current Indian market listings)
- Prices exclude GST and transportation unless stated otherwise.



JHARKHAND WATER ACCESS

DENSITY-BASED DEPLOYMENT PLAN

PEOPLE
RESOURCES
SUSTAINABLE
FUTURES

Community systems for dense settlements • Household systems for scattered mining-adjacent hamlets

Cleaner
Water
Stronger
Communities
A Healthier
Jharkhand



SAFE WATER
HEALTHIER LIVES



TARGETED SOLUTIONS
HIGHER IMPACT



MINING REGIONS
BRIGHTER TOMORROW

OUR CLASSIFICATION RULE

POPULATION
DENSITY
(persons/km²)

SETTLEMENT
PATTERN

WATER
MODEL

≥ 650

High density
(urban/clustered)

**COMMUNITY /
CENTRALIZED**

350 – 649

Moderate density
(semi-urban)

HYBRID
(both models)

< 350

Low density
(rural/scattered)

**HOUSEHOLD /
DECENTRALIZED**

i District-level screening only; final village/hamlet selection requires block/village verification.

JHARKHAND DISTRICT CLASSIFICATION

(Selected mining-intensive districts)

S. No.	District	Population Density (persons/km ²) (Census 2011)	Recommended Model
1	Dhanbad	1,316	Community
2	Bokaro	716	Community
3	Ramgarh	708	Community
4	East Singhbhum	648	Hybrid
5	Godda	622	Hybrid
6	Giridih	497	Hybrid
7	Pakur	498	Hybrid
8	Hazaribagh	488	Hybrid
9	Chatra	275	Household
10	Koderma	282	Household
11	Latehar	200	Household
12	West Singhbhum	209	Household

COMMUNITY / CENTRALIZED SYSTEM

For high-density settlements (≥ 650 persons/km²)



Applicable Districts
Dhanbad / Bokaro / Ramgarh

Examples

- Jharia – Katras – Sindri
- Bokaro steel/coal township clusters
- Dense Ramgarh coal-belt settlements

- ✓ Clustered households and shorter network distances make shared treatment, storage and distribution more economical.
- ✓ Suitable for towns, mining colonies and dense rural clusters.

HOUSEHOLD / DECENTRALIZED SYSTEM

For low-density settlements (< 350 persons/km²)



Applicable Districts
West Singhbhum / Latehar /
Chatra / Koderma

Example
Noamundi – Chiria – Manoharpur
mining belt

- ✓ Scattered hamlets and long pipeline distances make centralized systems uneconomical.
- ✓ Household or mini-solar decentralized systems are more feasible.



HYBRID APPROACH (350 – 649 persons/km²)

Applicable Districts: East Singhbhum, Godda, Giridih, Pakur, Hazaribagh

Use a mix of community systems in larger settlements and household units in scattered hamlets.

ILLUSTRATIVE PILOT & BUDGET REQUIREMENT



10,000 household units
(₹ 15,000 per unit)

= ₹ 15.0 Crore



20 community systems
(₹ 7 lakh per system)

= ₹ 1.4 Crore

TOTAL (Illustrative Pilot)

≈ ₹ 16.4 Crore

Note: Unit costs are illustrative planning costs – not government-approved prices.

SHARE OF GOVERNMENT BUDGET



0.42%

Our pilot requires
≈ **0.42%** of
₹ 3,952 Crore

₹ 3,952 Crore = assured Jharkhand water & sanitation tied-grant allocation for 2021–22 to 2025–26 in the cited government framework.



WHY IT CAN BE FUNDED

- ✓ DMF / PMKKKY prioritises drinking water and sanitation in mining-affected areas.
- ✓ Ramgarh DMFT reports 75.83% allocated to drinking-water supply.



INVESTING IN
CLEAN WATER TODAY
BUILDS A HEALTHIER,
STRONGER JHARKHAND TOMORROW.

Sources: Census 2011 district density data; PIB / Jal Jeevan Mission; Jharkhand Economic Survey 2024–25; DMFT Ramgarh. Values are planning inputs and must be validated before deployment.

PEOPLE | PLANET | PROGRESS